UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
B.Sc. Engineering
Semester 5 (21 Batch) Examination

BM3122 MEDICAL IMAGING

Time allowed: *Two* hours November 2024

ADDITIONAL MATERIAL:

- An information sheet on common constants and physical properties is attached.

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 6 pages.
- This examination accounts for 50% of the module assessment. The total maximum mark attainable is 100. The marks assigned for each question & sections thereof are indicated in square brackets.
- This is a closed-book examination.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.

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Information Sheet

Avogadro's number $= 6.02 \times 10^{23}$
Boltzmann's constant: $k = 1.38 \times 10^{-23}$ J/K
Density of water: $\rho_{\text{water}} = 1,000$ kg/m^3
Gyro-magnetic ratio of protons: $\gamma = 267.52 \times 10^6$ rad s^{-1}T^{-1}
Plank's constant: $h = 6.63 \times 10^{-34}$ Js

Some basic relationships – *Symbols have their usual meanings*

De Broglie's relationship:	$\Delta E = hf$
For spins in a strong magnetic field:	$N_{\text{parallel}} - N_{\text{anti-parallel}} = N_{\text{total}} \dfrac{\gamma h B_0}{4\pi k T}$
Net magnetization for spins in a strong magnetic field:	$\mathbf{M}_0 = \dfrac{\gamma h}{4\pi} \left(N_{\text{parallel}} - N_{\text{anti-parallel}}\right)$
Image intensity of MRI pixel acquired using a Gradient Echo sequence:	$I(x,y) \propto \rho(x,y) \dfrac{\left(1 - e^{-\text{TR}/\text{T}_1}\right) \sin\alpha}{1 - e^{-\text{TR}/\text{T}_1} \cos\alpha} e^{-\frac{\text{TE}}{\text{T}_2^*}}$
Optimized tip angle for a given TR:	$\alpha_{\text{Ernst}} = \cos^{-1} e^{-\frac{\text{TR}}{\text{T}_1}}$
The intensity of an axial MRI slice acquired using Spin Echo sequences:	$I(x,y) \propto \rho(x,y) \left(1 - e^{-\frac{\text{TR}}{\text{T}_1}}\right) e^{-\frac{\text{TE}}{\text{T}_2^*}}$
Radioactivity of a radioactive isotope:	$Q = \dfrac{dN}{dt} = \lambda N$
The number of radioactive nuclei at time, t:	$N(t) = N(t=0) e^{-\lambda t}$
The effective radiotracer half-life:	$\tau_{1/2,\text{eff}} = \dfrac{\tau_{1/2} \cdot \tau_{1/2,\text{bio}}}{\tau_{1/2} + \tau_{1/2,\text{bio}}}$
Gaussian line spread function:	$LSF(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\dfrac{(x - x_0)^2}{2\sigma^2}\right)$

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Question 1.

(a) Different imaging modalities are used to view the interior of the body in order to diagnose, monitor, or treat medical conditions.

i. Name **three** imaging modalities that use ionizing radiation and **three** non ionizing modalities. [3]

ii. For each imaging modality in (a) i., state one factor that affects its spatial resolution. [3]

iii. Line Spread Function (LSF) is used to measure the spatial resolution of an imaging system. For many imaging techniques, LSF is well-approximated by a Gaussian function. Starting from the Gaussian function, derive an expression for FWHM spatial resolution in terms of standard deviation. [4]

(b) A small amount of a radioactive material called a radiotracer is injected to the patient in nuclear medicine imaging.

i. State **four** desired properties of radiotracers used for SPECT imaging. [4]

ii. Give **three** factors for higher SNR of PET compared to SPECT. [3]

iii. What is the main limitation of PET imaging? [2]

iv. Two patients undergo nuclear medicine scans. One patient receives a dose of radiotracer A and the other radiotracer B. The half-lifes of A and B are 6 hours and 24 hours respectively. If the administered dose of radiotracer A is three times that of radiotracer B, and the biological half-lives of A and B are 6 and 12 hours respectively, at what time is the radioactivity in the body the same for the two patients? [6]

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Question 2.

(a) A radiography image is a transmission image showing the amount of X-ray attenuation along a path through the body.

i. State **two** classes of X-ray contrast agents with their main application. [4]

ii. State **three** main mechanisms of X-ray interactions with matter. [3]

iii. A bar phantom of thickness 0.4 cm is uniformly irradiated by monoenergetic X-ray photons, and a screen is placed behind the phantom to detect the X-ray photons. The bars of the phantom are made from a material that has an HVL of 0.1 cm. In the space between the bars, X-ray photons pass without attenuation. Assuming the intensity of the image is proportional to the X-ray photons that hit the screen in a unit area, what is the contrast of the resulting image? [5]

(b) Computed Tomography (CT) systems reconstruct a two-dimensional image slice from a series of one-dimensional X-ray projections.

i. State **three** advantages of CT imaging over projection radiography. [3]

ii. State the central section theorem. [4]

iii. Figure Q2 shows projections collected to create a 3×3 CT image. Reconstruct the image using Algebraic Reconstruction Technique (ART). Intermediate steps of calculations should be mentioned. [6]

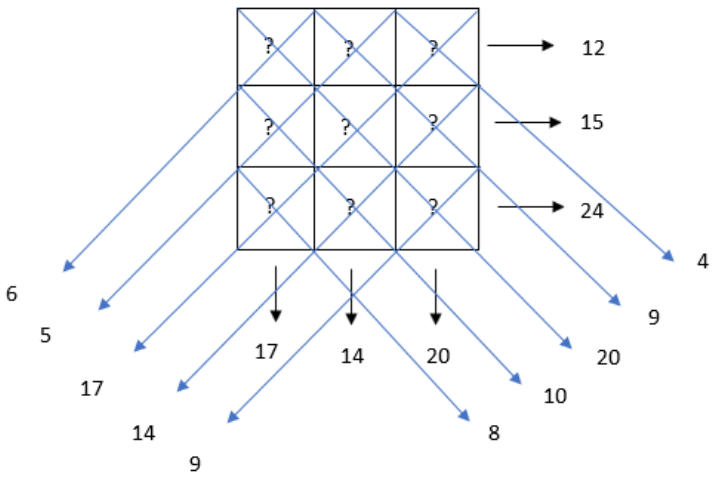


Figure Q2: CT projections

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Question 3.

(a) Magnetic Resonance Imaging (MRI) uses differences in magnetic properties of atomic nuclei to generate a spatial map of the hydrogen nuclei in different tissues.

i. Give **three** factors that can decide the intensity of an MRI image. [3]

ii. State **four** advantages of MRI over other imaging modalities. [2]

(b) The Gradient Echo (GE) pulse sequence is the most simplest pulse sequence used for MRI image acquisition.

i. Sketch a timing diagram of the basic GE pulse sequence. [5]

ii. Show how the pulse sequence in (b) i. can be modified to acquire multiple slices without increasing the acquisition time. [4]

iii. Explain briefy the tip angle (α) used in MRI acquisition. [2]

iv. An MRI of a uniform tissue phantom is acquired using GE pulse sequence with repetition time, TR = 1.5 T_1. What is the tip angle needed for the RF pulse to maximize the SNR? [2]

v. Calculate the percentage reduction of the SNR if a 90^0 tip angle is used instead of the optimum angle in (b) iv. [7]

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Question 4.

(a) Medical ultrasound is the second most commonly used imaging modality in clinical imaging.

i. State **four** advantages of ultrasound imaging. [2]

ii. State **three** common types of artifacts in clinical ultrasound and describe the cause of each. [6]

iii. State **two** methods to reduce one type of artifact in a (ii). [3]

iv. A linear array transducer has 256 elements where 8 non-overlapping elements work as an individual unit and fired sequentially to create an image. It is used to image an object with 50 cm depth. How many scan lines can be acquired in each frame? [2]

v. What is the frame rate in b (iv) assuming the speed of the sound in the object is 1540 m/s. [4]

vi. How could you increase the frame rate? [2]

(b) Optical Coherence Tomography (OCT) is an imaging technique very similar to clinical ultrasound.

i. Give **two** medical applications of OCT. [2]

ii. Describe the principle of operation of OCT. [4]

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End of Paper.